TOP-Q — Day 3 worked notes

A fair topological-vs-trivial comparison, disorder models, and the real hardware

PH 5110 term paper — Kotapati Kishore (EE26E016)

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Labels: [Derived] [Verified-by-test] [Simulated] [Literature] [Hypothesis]. Units: all b, κ, $\delta\beta$ in units of κ_2; baseline $r = \kappa_1/\kappa_2 = 1/3$ (decision D2); $M = 41$ guides. **Schedule:** Day 3 was planned for 24 Oct and is done on 25 Oct; the one-day slip is absorbed by the Day 7 buffer.

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1 Why the comparison must be designed, not picked

The research question compares a *topological* mode with an *ordinary* one. But an ordinary defect can be made as tightly or as loosely confined as we like. If we pick a loosely confined one, it will look fragile; if we pick a tightly confined one, it will look robust. Either way the "result" would be decided by our choice, not by physics (risk R7). So we fix **one rule for matching** before running any Monte Carlo, and write down why.

2 Array T: the topological interface (from the literature)

Geometry. Following Blanco-Redondo *et al.* (2016) [Literature], the topological structure is a **"long-long" domain wall**: the centre guide has a *wide* gap (weak coupling κ_1) on *both* sides; moving outward the couplings alternate $\kappa_2, \kappa_1, \kappa_2, \ldots$ (Fig. 2a). It is two SSH chains glued back to back, mirror images of each other.

Mode [Derived]. Exactly as on Day 2 but in both directions: on the centre's sublattice the amplitude is multiplied by $-r$ per cell; on the other sublattice it is zero:

$$\psi_{2j} = \psi_0(-r)^{|j|}, \qquad \psi_{2j+1} = 0, \qquad b_T = 0 \text{ exactly.}$$

The array has an odd number of guides and only A–B couplings, so the Day-2 counting argument guarantees this zero mode for any coupling values. Numerically $|b_T| = 2 \times 10^{-16}$ and the eigenvector matches the formula to 10^{-12} [Verified-by-test].

In simple words: light put in the centre guide sits there because both its neighbours are only weakly attached. The little light that leaks out skips every second guide.

3 A surprise: the "short-short" wall is not trivial in our model

Blanco-Redondo *et al.* also built a "short-short" structure (narrow gaps next to the centre) and reported *trivial* defect modes there, which they attributed to the three closely spaced guides forming a region of higher effective index [Literature].

We built the same wall with couplings only. Result [Simulated; Verified-by-test]: it is chiral-symmetric and has three near-zero modes — one on the two guides *next to* the centre (the centre guide itself stays dark, power $< 10^{-9}$) and two at the array ends. In a nearest-neighbour coupling model, a short-short wall is just the *other* kind of SSH domain wall, which also carries a mid-gap state.

What this means. The trivial behaviour they saw comes from physics *not in our model*: the extra effective index of closely packed guides is an **on-site** effect (the same kind as Limitation L1). So the paper's short-short structure cannot serve as our trivial reference unless we add on-site terms whose size we do not know. **Decision D5:** our trivial reference is an explicit, controlled on-site defect (array D below).

Viva point. "Topological vs trivial" is a property of a *model*. The same physical structure can look different in a coupling-only model and in the real device. Say which model you mean.

4 Array D: a conventional defect with matched confinement

Geometry. Same number of guides, every coupling equal to the mean $\bar{\kappa} = (\kappa_1 + \kappa_2)/2 = 2/3$, and the centre guide's propagation constant raised by Δ (a slightly wider guide, Fig. 2b).

4.1 Bound mode [Derived]

Try $\psi_n = q^{|n|}$ with $0 < q < 1$. Away from the centre, $b\,\psi_n = \bar{\kappa}(\psi_{n-1} + \psi_{n+1})$ gives $b = \bar{\kappa}(q + 1/q)$. At the centre, $b\,\psi_0 = \Delta\psi_0 + 2\bar{\kappa}q$ gives $b = \Delta + 2\bar{\kappa}q$. Equating:

$$\bar{\kappa}\Big(\frac{1}{q} - q\Big) = \Delta \;\Rightarrow\; q = \frac{\sqrt{\Delta^2 + 4\bar{\kappa}^2} - \Delta}{2\bar{\kappa}}, \qquad b_D = \sqrt{\Delta^2 + 4\bar{\kappa}^2}.$$

The mode exists for any $\Delta > 0$ and sits *above* the band of the uniform array ($|b| \le 2\bar{\kappa}$). Nothing pins b_D: it moves whenever Δ or $\bar{\kappa}$ moves.

4.2 The matching rule: equal IPR [Derived]

Inverse participation ratio $\mathrm{IPR} = \sum |\psi_n|^4$ (normalised mode): 1 = all light in one guide. For power $p_n \propto q^{2|n|}$:

$$\mathrm{IPR}(q) = \frac{(1+q^4)(1-q^2)}{(1+q^2)^3}.$$

The T mode has *the same formula with* $q \to r$ (its zeros do not change the sums). So equal IPR $\Leftrightarrow q = r$, giving

$$\boxed{\Delta = \bar{\kappa}\Big(\frac{1}{r} - r\Big) = \tfrac{16}{9}\kappa_2, \qquad b_D = \tfrac{20}{9}\kappa_2, \qquad \mathrm{IPR}_T = \mathrm{IPR}_D = 0.656.}$$

Why IPR, and not something else? Section 6 shows that, to first order, an on-site error spreads b by $\sigma_\beta\sqrt{\mathrm{IPR}}$ for *any* mode. Matching IPR therefore makes T and D **equally exposed to on-site disorder by construction**. Any difference we then see under coupling disorder is due to the structure (topology), not to confinement. **Honest limitations of D** (record as L6): D sits in a gapless uniform array, T in a gapped one; D's mode is above the band, T's is mid-gap; IPR matching makes D decay per *guide* while T decays per *two guides*. A sensitivity check with another matching rule (equal decay length per guide) is an optional Day-4 item.

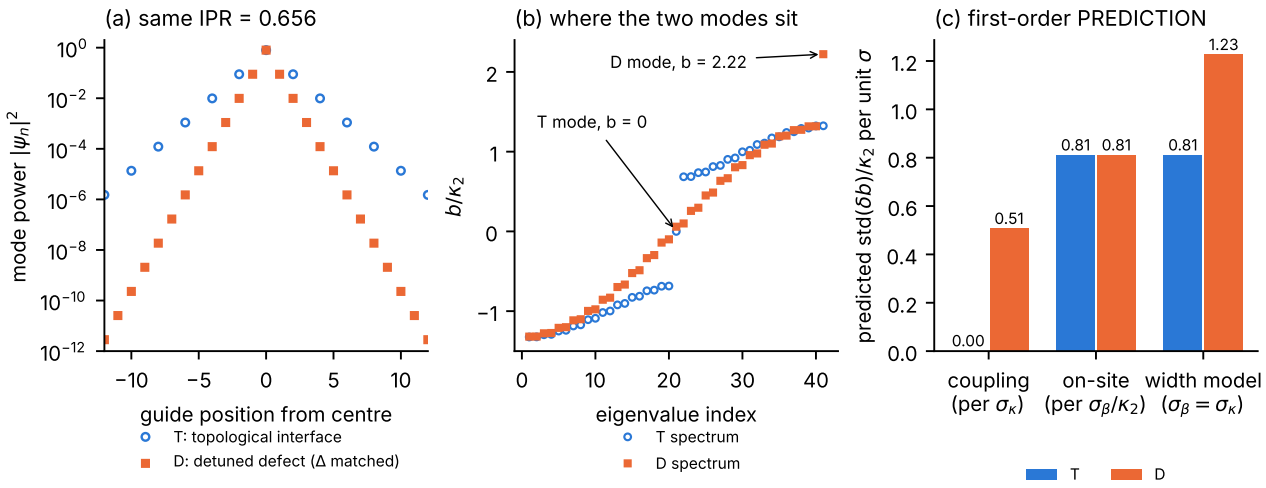


Figure 1: Generated by `scripts/day3_design.py` from `data/day3_design.npz`. (a) Mode power of T and D with equal IPR. (b) Spectra: T mode at $b = 0$ in the gap; D mode above the band at $b = 2.22\kappa_2$. (c) First-order **predictions** (not Monte Carlo) of the spread of b per unit disorder strength.

5 Disorder models (what each one means physically)

5.1 Coupling disorder: log-normal, and why [Derived]

On Day 1: $\kappa = \kappa_0 e^{-\gamma g}$. If the gap error δg is Gaussian, then $\ln\kappa$ changes by $-\gamma\,\delta g$, which is Gaussian. So $\kappa \to \kappa\, e^{\sigma_\kappa g}$, $g \sim \mathcal{N}(0,1)$, $\sigma_\kappa = \gamma\sigma_g$, is *exactly* what a Gaussian gap error produces, and κ can never go negative. Keeps chiral symmetry. [Verified-by-test: positive, $\langle\ln\rangle = 0$, std $= \sigma_\kappa$]

5.2 On-site disorder

$\delta\beta_n = \sigma_\beta g_n$: each guide's propagation constant is off by a random amount (width or height error). Breaks chiral symmetry.

5.3 Width model: the realistic, mixed case

One width error δw_n per guide changes *two* things at once:

$$\delta\beta_n = \sigma_\beta\,\delta w_n, \qquad \kappa_{n,n+1} \to \kappa_{n,n+1}\exp\!\left[\sigma_\kappa \frac{\delta w_n + \delta w_{n+1}}{2}\right].$$

Sign logic: if the guide centres are fixed by the mask, a wider guide leaves a narrower gap, so the coupling grows. Neglected: the direct change of κ with how tightly a wider mode is confined (**L5**). The ratio $\sigma_\beta/\sigma_\kappa$ depends on the waveguide and **is not known yet** (needs a mode solver); Day 4 scans it rather than guessing it. [Verified-by-test: correlation between a guide's $\delta\beta$ and its couplings > 0.6]

5.4 Global shift

One common offset on all guides (wafer-scale thickness error). Breaks chiral symmetry but adds the same phase to everything, so it cancels in relative-phase measurements (Day 2, question 2).

5.5 Reproducible random numbers

Each Monte Carlo sample s gets its own generator from `SeedSequence([seed, s])`. The same seed and sample number always give the same numbers, and the result does not depend on the order in which samples are run [Verified-by-test].

6 First-order predictions (to be tested on Day 4)

For a small change δH, the eigenvalue moves by $\delta b \approx \langle\psi|\delta H|\psi\rangle$ (first-order perturbation theory) [Derived; Verified-by-test against exact diagonalisation, error $< 10^{-10}$]:

$$\delta b \approx \sum_n \psi_n^2\, \delta\beta_n + 2\sum_n \psi_n \psi_{n+1}\, \delta\kappa_n.$$

- **T under coupling disorder:** $\psi_n\psi_{n+1} = 0$ for every bond (one sublattice is empty), so the first-order shift is *zero*. (Day 2 showed it is zero to *all* orders.)
- **On-site disorder, any mode:** $\mathrm{std}(\delta b) = \sigma_\beta\sqrt{\sum \psi_n^4} = \sigma_\beta\sqrt{\mathrm{IPR}}$. Equal for T and D by design.
- **Width model:** T feels only the on-site part; D feels both, and they add because they are correlated.

Predicted $\mathrm{std}(\delta b)/\kappa_2$ per unit σ	coupling (σ_κ)	on-site (σ_β)	width ($\sigma_\beta = \sigma_\kappa$)
T (topological interface)	0.00	0.81	0.81
D (matched defect)	0.51	0.81	1.23

Hypotheses for Day 4. H3: under coupling disorder T stays at $b = 0$ while D spreads as $0.51\sigma_\kappa$ for small σ. **H4**: under on-site disorder T and D spread *equally* ($0.81\sigma_\beta$): topology gives *no* protection here. **H5**: under the width model T is better than D only by removing the coupling part; how much better depends on the unknown ratio $\sigma_\beta/\sigma_\kappa$.

Big picture. If H4 holds, the first honest conclusion of TOP-Q is already visible: topological protection removes *one* error channel (coupling errors) and does nothing for another (propagation-constant errors). Whether real fabrication errors are mostly one or the other decides how useful the protection is.

7 The real hardware

What is ours and what is not. We have **no** fabricated chip, laser or camera. Every number in TOP-Q is a simulation. The hardware below is the *published* platform that our model describes.

7.1 How the real device works (step by step)

1. **Chip.** A silicon-on-insulator wafer: a 220 nm silicon layer on a buried oxide layer on a silicon substrate. Deep-UV lithography (at IMEC in the paper) and etching leave silicon strips 450 nm wide — these are the waveguides. Air above acts as cladding.
2. **The design is only in the gaps.** All guides are identical; the SSH pattern is written by alternating the gaps (166 and 294 nm). This is why gap errors → coupling disorder, and width errors → both (Section 5.3).
3. **Input.** A 1550 nm laser is coupled into the centre guide.
4. **Propagation.** 500 μm of propagation; the paper notes this should be at least about 10 coupling lengths.
5. **Output.** The light leaving the end of the array is imaged on a near-infrared camera, and the power in each guide is measured with a fibre connected to a power meter.
6. **Result reported.** The long-long (topological) structure kept light in the centre guide with the same output over an 8 nm wavelength range; the short-short structure showed beating between two modes that changed with length and wavelength [Literature].

7.2 Published hardware images to look at and cite

Real photographs and SEM images are owned by their publishers, and my workspace cannot download them, so they are **not** pasted into these notes. They were shown in our chat, and you can find them here:

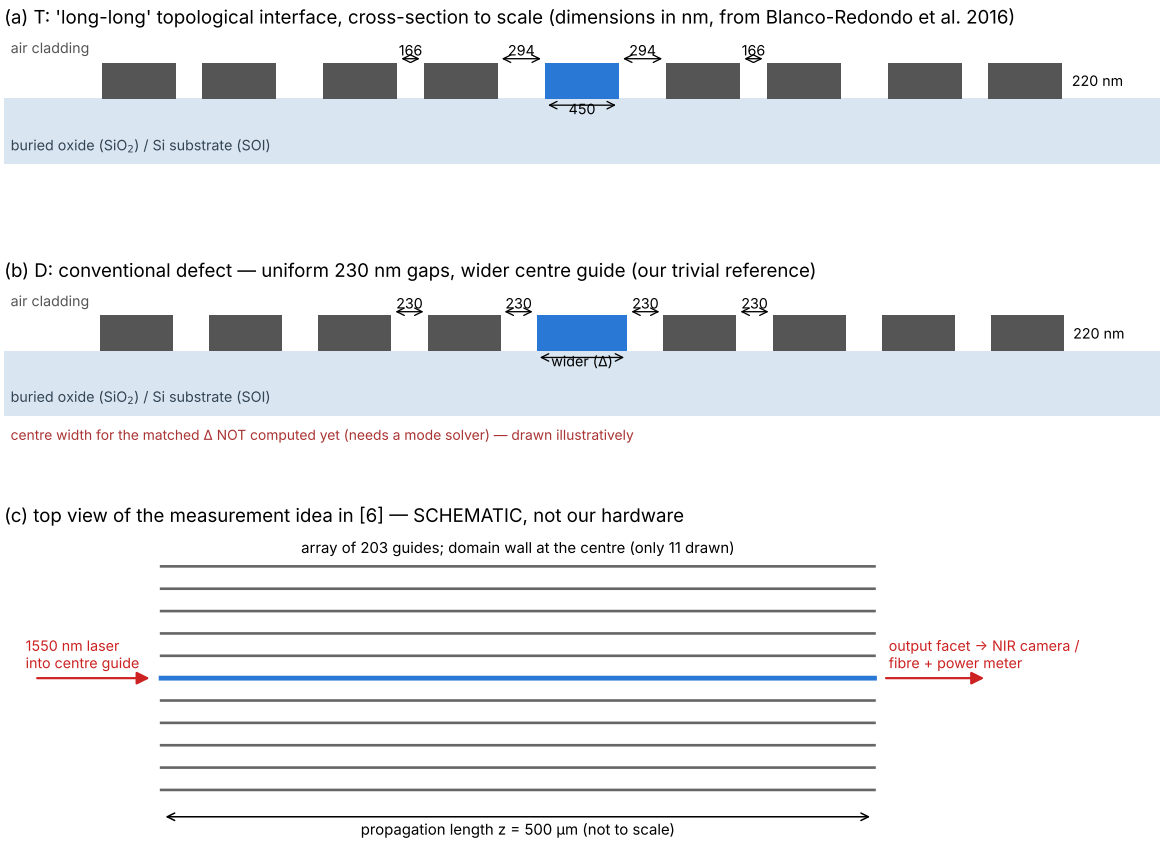


Figure 2: **Schematic drawn by us** (not a photograph). Cross-sections to scale with dimensions from Blanco-Redondo *et al.* (2016): Si waveguides 450×220 nm on silicon-on-insulator, air cladding, 203 guides, 500 μm long, $\lambda = 1550$ nm; long-long gaps 294/166 nm. (b) Our trivial reference D; the wider centre width is illustrative because the width that gives the matched Δ has not been computed. (c) Measurement idea: light into the centre guide, output imaged with a near-infrared camera and measured with a fibre and power meter.

Source	What to look for
Blanco-Redondo *et al.*, PRL **116**, 163901 (2016)	device images of the silicon arrays and measured output patterns (the platform we model)
Review "Topological silicon photonics", *Advanced Photonics Research* (2021) — *verify authors and DOI*	SEM images of many silicon topological devices
Review "Integrated topological photonics in one dimension" (Taylor & Francis) — *verify authors, journal and DOI*	1-D SSH chips, fabrication and measurement setups
Femtosecond-laser-written waveguides in fused silica (several groups)	a second real SSH platform: waveguides written inside glass, with fluorescence imaging of light inside the array

Rule for the paper: any real photo must say "Reproduced from [ref]" and should come from an open-access (CC-BY) source or have permission. Never caption it as our own device.

8 Code and tests

New: `topq/arrays.py` (T, D and SS arrays, matched Δ, analytic modes), `topq/disorder.py` (coupling, on-site, width and global disorder; reproducible generators; first-order predictions). `tests/test_arrays_disorder.py`: 16 new tests, including a 3000-sample Monte Carlo check of the width-model prediction (agreement within 5%). Whole suite: **70 passed**.

9 Self-test (answer without notes)

1. Why would picking an arbitrary trivial defect make the comparison meaningless?
2. Derive $b_D = \sqrt{\Delta^2 + 4\bar{\kappa}^2}$ in three lines.
3. Why does matching IPR make T and D equally sensitive to on-site errors?
4. Why is a Gaussian gap error the same thing as log-normal coupling disorder?
5. In the width model, why does a wider guide *increase* its neighbours' couplings?
6. The paper calls the short-short modes trivial, but our model finds a mid-gap state. Who is wrong?